

Exp.No-4: OFDMA Resource Block Allocation Date

CODE IMPLEMENTATION:

```
OFDMA_Resource_Block_Allocation.m ×
/MATLAB Drive/OFDMA_Resource_Block_Allocation.m
1  clc;
2  clear;
3  close all;
4
5  total_bandwidth = 20;
6  rb_bandwidth = 0.18;
7  allocated_RBs = 80;
8  spectral_efficiency = 4;
9
10 total_RBs = floor(total_bandwidth/rb_bandwidth);
11 utilization = (allocated_RBs/total_RBs)*100;
12 throughput = allocated_RBs*rb_bandwidth*spectral_efficiency;
13
14 fprintf('===== OFDMA RESOURCE BLOCK ALLOCATION =====\n\n');
15 fprintf('Total Bandwidth = %.0f MHz\n',total_bandwidth);
16 fprintf('Total Resource Blocks = %d\n',total_RBs);
17 fprintf('Allocated Resource Blocks = %d\n',allocated_RBs);
18 fprintf('RB Utilization = %.2f %%\n',utilization);
19 fprintf('User Throughput = %.2f Mbps\n',throughput);
20
21 figure;
22
23 subplot(2,2,1);
24 bar(total_RBs);
25 grid on;
26 title('Objective 1');
27 label('OFDMA System');
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

```
Total Resource Blocks = 111
Allocated Resource Blocks = 80
RB Utilization = 72.07 %
User Throughput = 57.60 Mbps
>>
```

OUTPUT:

